

• Section A -

1) What is a string in python?

→: String is a sequence of character and we used single (') and double (") quotes.

2) Which symbol is used to create a tuple?

→: Parentheses ()

3) Which python data type stores unique values?

→: Set()

4) Write the syntax to create an empty dictionary.

→: Syntax →

dict = {}

5) Which loop is generally used to iterate over a sequence?

→: for loop.

6) Which keyword is used to exit a loop immediately?

→: break keyword

7) Can a list contain duplicate values?

→: yes...

8) Can a tuple be modified after it is created?

→: No, because it is immutable.

9) What is the index of the first ele of a python list?

→: 0.

10) Which function is used to find the no. of ele's in a string,

→: len()

list, tuple, set, or dictionary.

• Section B - 2 Marks Que.

1). Differentiate betⁿ list and tuple.

→:

list	tuple
------	-------

- | | |
|----------------------------------|------------------------------------|
| 1) mutable | Immutable |
| 2) [] | () |
| 3) it can store duplicate values | it can also store duplicate values |
| 4) list start from index 0. | tuple it start from index 0. |

2). Diff. betⁿ set and dictionary.

→:

set	Dictionary
-----	------------

- | | |
|---|---|
| 1) mutable | mutable |
| 2) {} | {} |
| 3) set stores unique values | Dictionary stores unique key. |
| 4) set is a collect ⁿ of unique elements | Dictionary is collect ⁿ of key-value pairs |

3) Write a python program to count the number of characters in a string.

→:

```
string = "python"  
print(len(string))
```

4) Write a python program using a for loop to print numbers from 1 to 10.

→ for i in range(1,11):
 print(i)

5) Write a python program using a while loop to print numbers from 10 to 1.

→ i = 10
while i >= 1:
 print(i)
 i = i - 1

• Section C - 5 M. Questions -

1) String -

Consider: text = "Python Programming".

Write a python statements to:

a) print the complete string.

→
• text = "Python Programming"
 print(text)

b) print the first character.

→
• text = "Python Programming"
 print(text[0])

c) print the last character.

• text = "Python Programming"
 print(text[-1])

d) print "python" using slicing.

→

```
text = "python"
```

```
print(text[0:5:2])
```

e) Count the no. of characters in the string.

→

```
text = "python programming"
```

```
print(len(text))
```

2) List - 5 m.

Create the following list:

```
Numbers = [10, 20, 30, 40, 50]
```

write python statements to:

a) Add 60 to the list.

→

```
Numbers = [10, 20, 30, 40, 50]
```

```
Numbers.append(60)
```

```
print(Numbers)
```

b) Remove 20

→

```
Numbers = [10, 20, 30, 40, 50]
```

```
Numbers.remove(20)
```

```
print(Numbers)
```

3) change 30 to 35.

d) print the first 3 elements.

→ numbers = [10, 20, 30, 40, 50]
print(numbers[:3])

e) print the length of the list.

→ print(len(numbers))

3] Tuple - SM

Create the following tuple:

fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

write python statements to:

a) print the tuple.

→ fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")
print(fruits)

b) print "Mango".

→ print(fruits[2])

c) print the first two ele.

→ print(fruits[:2])

d) find the length of the tuple.

→ print(len(fruits))

e) check whether "Apple" exists in the tuple.

→

4) Set.

Consider:

numbers = {10, 20, 30, 20, 40, 10, 50}

a) Ans. the following:

a). What happens to duplicate values in a set?

→: Duplicate values in a set are automatically removed.

b). Add 60 to the set.

→: we use the add() method.

c). Remove 30 from the set.

→: use remove() method
- set.remove(30)

d). find the number of ele. in the set.

→: Use the len() fun.

5) Dictionary -

create the following dictionary:

Student = {"name": "Rahul", "age": 22, "course": "Python",
"city": "Delhi"}

write python statements to:

a) print the student's name.

→: student = {"name": "Rahul", "age": 22, "course": "Python", "city": "Delhi"}

Print(student["name"])

b) print the student's age.

→: `print(student["age"])`

c) Display all keys of the dictionary.

→: `print(student.keys())`
`print(num.keys())`

* Loops -

A] for loop -

a) Write a python program using a for loop to print no's from 1 to 10.

→: `for i in range(1, 11):`
`print(i)`

b) Write a python program using a for loop to print all even no's from 1 to 20.

→: `for i in range(1, 20):`
`if i % 2 == 0:`
`print(i)`

B] While loop -

a) Write a python program using a while loop to print numbers from 1 to 10.

→.

```
→: i = 1
while i <= 10:
    print(i)
    i = i + 1
```

b) Write a python program using a while loop to calculate the sum of no's from 1 to 10.

```
→: i = 1
sum = 0
while i <= 10:
    sum = sum + i
    i = i + 1
    print(sum)
```

c) Write a program using a while loop to print no's from 10 to 1.

```
→: i = 10
while i >= 1:
    print(i)
    i = i - 1
```